Rayyan Moosani

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Education

University of Waterloo – Bachelor of Computer Science Waterloo, ON

Experience

Software Engineer May 2026 - Present

XORBIX

- Cut validation time **50x** to **442K rows/sec** via chunked vectorized **pandas** boolean-mask engine
- Slashed **Flask REST API** latency **5.5x** (3.1s to 0.56s) by collapsing 301 **N+1** queries into 3 bulk reads
- Reduced AI extraction time by **95%** (6 min to 5s) in a **C/.NET WPF** app via real-time data streaming
- Generated **200+** files of **.NET/React** apps, cutting setup effort **85%**, via self-made 6-step codegen application

Software Developer Sep 2025 – Dec 2025

IICUW

- Optimized frontend performance using ES6+ refactoring, reducing bundle size by **20%** and improving load time
- Engineered **Python automation tools** to orchestrate internal workflows, boosting efficiency by **35%**.
- Built real-time **React** features using event-driven architecture, increasing engagement for **100+ users**

Software Automation Developer Aug 2024 - Sep 2024

SCITRA

- Cut reporting latency by **40%** across **30+ factory lines** via real-time **Telegram & Sheets API** integration.
- Engineered **Unity** and **C#** digital twins with **IoT** data, improving predictive maintenance accuracy by **30%**
- Automated fault logging using **Python** pipelines, accelerating engineering resolution time by **25%**

IT Intern Jun 2024 - Jul 2024

AGMC

- Digitized workflows using **MS Forms** and **SharePoint**, reducing manual processing effort by **30%**
- Migrated legacy systems to **M365**, improving data accessibility and retrieval speed by **40%**
- Resolved **50+ IT tickets** using enterprise tools, improving system reliability and user productivity

Projects

StoryVerse | *TypeScript, Node.js, Express, React, PostgreSQL, Drizzle ORM, FFmpeg, OpenAPI, Zod*

- Shipped a **portal** where viewers watch AI trailers made by authors and **vote to greenlight** them for **funding**.
- Built **AI pipeline** turning scripts into trailers and shorts across **7+ tiers** of **20+ models**.
- Cut generation cost **85%** by building an **editable**, **human-in-the-loop** pipeline state machine.
- Raised **character consistency ∼2/9 to 9/9** via **prompt-injection** layer locking descriptions.
- Synced multi-scene **narration** and **lip-sync** to video via **FFmpeg** with **R2**-backed storage.

Circuit Guard | *Python, FastAPI, Next.js, React Three Fiber* **Winner**, FormulaTechHacks 2026

- Built an **AI Model** that spots F1 crashes and tells officials which flag to show, quoting the F1 rulebook.
- Mapped **600+** real race incidents to show which corners endanger fans, with a low-cost fix for each.
- Pinpointed the 3 corners behind **47%** of serious crashes so track owners can negotiate cheaper **insurance**.

Applicant Zero | *Python, LangGraph, Claude, Playwright, FastAPI, Telegram Bot API, Linux, Tailscale*

- Architected **8 AI agents** on a **LangGraph** state machine, achieving **a tailored application from 1 job link**
- Grounded **Claude** in verified facts, achieving **tailored resumes and cover letters with zero fabrication**
- Built **self-healing loops** that re-route on ATS and form errors, achieving **autonomous recovery mid-run**
- Added a **Telegram** + **FastAPI** approval gate on every submission, achieving **zero sends without my sign-off**
- Deployed as an **always-on Linux service** over **Tailscale**, achieving **24/7 runs controlled from my phone**

Technical Skills

Languages: Python, C#, TypeScript/JavaScript, SQL, C, Racket, XAML, HTML5, CSS3.
AI/ML: LLMs, RAG pipelines, Anthropic Claude API, OpenAI API, Ollama, LM Studio, Claude Code, prompt engineering, Whisper, Pandas, NumPy, PHOIBLE, Selenium.
Frameworks, Infrastructure & APIs: .NET 8, ASP.NET Core, WPF (MVVM), EF Core, FastAPI, Flask, React, Vite, Node.js, Express, TailwindCSS, Docker, PostgreSQL (pgvector), SQLite, Supabase, MongoDB, REST APIs, WebSockets, OAuth 2.0, xUnit, Azure DevOps, CI/CD (Jenkins), Git/GitHub.